

外

平成16年度 筑波大学

大学院修士課程 理工学研究科

入学試験問題

大学院博士課程 システム情報工学研究科(3 専攻)

外国語

えいご
英語

以下の二つの問題いずれにも答えなさい。Problem 1 と Problem 2 を別々の解答用紙に解答すること。
Answer both Problem 1 and 2 in separate answer sheets, respectively.

Problem 1

WEB ATTACK

Hackers take down Yahoo!

WASHINGTON (AP) (A)Computer vandals using a common electronic attack overwhelmed Yahoo!, the most popular site on the Internet, and rendered the flagship Web directory inaccessible for hours.

Yahoo! Inc. spokeswoman Diane Hunt said the company, worth roughly \$93 billion, was the victim of hackers flooding its equipment with repeated electronic requests. The vandals did not gain access into the Web site's computers, she said.

The directory was inaccessible for much of Monday.

①The technique, called a denial of service attack, is similar to pranksters repeatedly dialing a company's telephone number to block all other incoming calls.

The failure drew renewed attention to the risks facing the fledgling world of electronic commerce, where hackers can shut down even the largest online stores.

"It basically says nobody is safe, if Yahoo! can be taken down with all the resources behind them," said Elias Levy, chief technology officer, at San Mateo, California-based Security Focus Com.

Hunt said technicians determined that a flood of data requests coming from different computers on the Internet had overwhelmed its routers, which help direct traffic for the Web site.

Technicians ultimately were able to identify the type of data and filter it out, which restored service.

Media Metrix, which tracks visitors to Web sites, said www.yahoo.com was the most popular Internet site for the month of December, the latest figures available, when it recorded 36 million visitors.

She declined to say whether Yahoo! had contacted the FBI, which coincidentally warned Web sites last month about a specific type of denial of service attack.

"Our first priority has been identifying what was happening and then installing the filters to enable our users to access our services," Hunt said.

The FBI suffered a similar attack on its own Web site last year. It made the site inaccessible for days in April 1999 after hackers overwhelmed its Internet computers also using a denial of service attack.

In its new advisory, ②the FBI's National Infrastructure Protection Center said it was "highly concerned about the scale and significance of these reports" and said vulnerabilities were "wide-spread, well-known and readily accessible on most networked systems."

Yahoo! said the problems began Monday afternoon. The problem also prevented the company's customers from accessing free e-mail accounts through the Web site, but it didn't otherwise affect the flow of mail.

Also, the problem did not affect customers' home pages on Geocities, which is owned by Yahoo! (The Japan Times, Wednesday, February 9, 2000)

1. 下線部①、②を適宜わかりやすく説明せよ。

Explain both the underlined part ① and ② in detail.

2. 上記の記事では下線 (A) に対して Yahoo! はどのような対策が取れると述べているか、簡潔にわかりやすく説明せよ。

Against the "common electronic attack" mentioned in (A), briefly explain what action that Yahoo! can take in this article.

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Problem 2

下記の英・日両記事を読み、下線部の日本語を参考に、英文記事の()部を英文で埋め、段落を完成させよ。適宜前後関係も考慮せよ。

Read both the following Japanese and English articles and fill out the blank part () in the English article to continue and conclude the paragraph referring to both the influences and applicabilities of Chomsky's study. You can also refer to the underlined Japanese sentences.

The science of linguistics has fully as many facets and fields as biology, and like biology, what may be called its 'modern era' can be traced to the 1950s. The decade that unveiled the structure of DNA also witnessed a revolution in linguistics led by Noam Chomsky, whose work radically diversified the field beyond its then-current focus on simply cataloguing the actual utterances of a language, to exploring the mechanisms by which they are produced. Seeking to identify the universals at the core of all languages, he posited a new, generative form of grammar, or set of syntactic rules, that would help to account for the immense creativity in the production of language that emerges so rapidly as individuals develop.

In pursuit of his 'universal grammar', Chomsky created waves that washed up on many scientific shores. Besides his profound influence on theoretical linguistics, his mathematical approach to the description of languages prompted a burst of development in formal language theory. This produced methods with widespread utility in computer science, from the specification and interpretation of computer languages to the fields of syntactic pattern recognition, natural language processing and speech understanding. The Chomsky hierarchy of language classes has proven especially durable as a means of stratifying formal languages according to their expressive power and resulting computational and mathematical complexity. ()

(NATURE VOL 420, David B.Searls "The language of genes"より)

言語学には、生物学と同じように、たくさんの分野があり、現代言語学が始まったのも現代生物学と同じく1950年代のことだった。その10年間にDNAの構造が明らかになったのだが、言語学ではノーム・チョムスキー (Noam Chomsky) が主導する革命的な進展が見られた。当時の言語学研究の焦点は、言語の実際の発話をカタログ化することに当てられていたが、チョムスキーの研究によって、言語学の対象は、発話のメカニズムにまで広がっていった。彼は、全ての言語の核心部分に普遍的なものを見つけ出そうと試み、新しい生成文法、または、統語論の規則のセットを提案した。それは、個人の発達過程にしたがって、急速に出現する発話生成の膨大な創造力を説明するのに役立つであろうと考えられている。

チョムスキーの「普遍文法」の探求は、さまざまな科学の最前線に影響を与えた。理論言語学に重大な影響を与えるのみならず、彼の言語の記述への数学的なアプローチは、形式言語理論の爆発的な発展を促した。それによって、コンピュータサイエンスにおいて、コンピュータ言語の仕様および解釈から、文法パターン認識、自然言語処理と音声理解の分野まで、広範囲に役立つ方法が生み出された。言語クラスのチョムスキー階層は、表現力と、コンピュータ的・数学的複雑さを基盤に、形式言語を階層化する方法として有用であることが証明された。チョムスキー(Chomsky)の影響は、認知科学、分析哲学と文芸批評にさえも及んだ。複数の研究分野を通じて共通しているのは、言語学からもたらされた分析技術だけでなく、言語学的感受性とも言うべきものがあれば、別の同じくらい複雑な研究分野についても解明を進められるということである。

認知科学 : cognitive science, 分析哲学 : analytic philosophy
文芸批評 : literary criticism
言語学的感受性 : linguistic sensibility

(NATURE VOL 420, David B.Searls "The language of genes"和訳版より)